

Lab 1 - ASP.NET Core MVC Basics

Objective

In this lab, you will learn how to:

- Create an ASP.NET Core MVC project.
 - Create an MVC Controller.
 - Pass data from the controller to a view.
 - Work with Query Parameters.
 - Display data using Razor Views.
 - Create a simple styled HTML table.
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Requirements

Step 1: Create an MVC Project

Create a new **ASP.NET Core Web App (Model-View-Controller)** project.

Project Name:

StudentManagement

Step 2: Create a Student Model

Create a model named:

Student

The model should contain the following properties:

- Id
 - Name
 - Age
 - Department
 - GPA
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Step 3: Create Sample Data

Inside the controller, create a list of at least **5 students** with different values.

Example:

- Ahmed
- Sara
- Mohamed
- Mariam
- Omar

You are free to choose the values.

Step 4: Create StudentController

Create a controller named:

StudentController

The controller should contain **two action methods**.

Action 1: Show All Students

Action Name:

Index

Requirements:

- Return a View.
- Display all students.
- The page must contain a styled HTML table.
- Display the following columns:

Id Name Age Department GPA

Style the table using CSS or Bootstrap.

Action 2: Show Student By Id

Action Name:

Details

Requirements:

- Accept an **Id** using **Query Parameters**.
- Search for the student in the list.
- If the student exists, display all of their information.
- If the student does not exist, display a suitable message such as:

Student Not Found

Query Parameter Example

Example URL:

`https://localhost:5001/Student/Details?id=3`

or

`https://localhost:5001/Student/Details?id=1`

In this example:

- Details is the action method.
- id is passed as a Query Parameter.
- The controller should retrieve and display the corresponding student.

Views

Create the following views:

Views

└── Student

 └── Index.cshtml

 └── Details.cshtml

Expected Output

Index Page

- Page title: **Students**
- Styled table
- Five student records displayed

Example:

Id	Name	Age	Department	GPA
1	Ahmed	21	Computer Science	3.4
2	Sara	22	Information Systems	3.8
...	...	...	...	...

Details Page

Display something similar to:

Student Details

Id: 3

Name: Mohamed

Age: 20

Department: Software Engineering

GPA: 3.7

If the student is not found:

Student Not Found

Bonus (Optional)

1. Add Bootstrap styling to both pages.
 2. Add a "Back to Students" button on the Details page.
 3. Highlight students with GPA greater than 3.5 using a different background color.
 4. Display the total number of students at the top of the Index page.
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Submission

Submit the complete Visual Studio solution containing:

- MVC Project
- Student Model
- StudentController
- Index View
- Details View

Ensure the project runs successfully without errors.